

InfantCryNet

Multimodal Clinical Inference Engine for Infant Cry Analysis

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Ethics Protocol: CHU-SBA-2025-017 | **Language:** Python 3.10+ | **License:** MIT

Multimodal Fusion

Conformal Safety

IRB Compliant

Explainable AI

Open Inference Engine

Abstract

InfantCryNet is an open-source, production-grade multimodal inference engine for the automated clinical interpretation of infant crying behaviour. The system integrates acoustic signals, biological context, and visual motor patterns through a novel **Reliability-Weighted Product-of-Experts (RW-PoE)** fusion algorithm, providing calibrated posterior distributions over diagnostic categories. The safety architecture is grounded in the **Bhattacharyya-Barrier** proof and **conformal prediction sets** that guarantee marginal coverage $\mathbb{P}(\hat{y} \in \mathcal{C}(x)) \geq 1 - \alpha$. Validated on clinical data collected under IRB protocol **CHU-SBA-2025-017**, the pipeline achieves a **34.8 percentage-point accuracy gain** over acoustic-only baselines.

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Project Overview

What This Repository Provides

InfantCryNet is a modular, software framework that any hospital or research group can deploy against their own ethically collected infant cry recordings.

Component	Status	Location
Acoustic feature extraction	Open Source	src/modules/acoustic/
Biological expert system (XAI)	Open Source	src/modules/biological/
Visual stochastic simulator	Open Source	src/modules/visual/
RW-PoE fusion engine	Open Source	src/core/
Conformal prediction wrapper	Open Source	src/core/
Raw clinical data	Restricted	CHU-SBA secure vault
Trained model weights	On Request	MTA required

Data Privacy & Ethical Compliance

Data Privacy & Ethical Compliance

IRB Protocol: CHU-SBA-2025-017. All clinical data are classified as protected health information under Algerian Law No. 18-11. This repository does not contain raw clinical material. Researchers must apply this engine to their own ethically collected data.

System Architecture

Acoustic Engine

The acoustic branch extracts MFCCs, spectral descriptors, and F_0 pitch contours.

- **Node A:** Binary classifier (Hungry). AUC = 0.5604.
- **Node B:** Binary classifier (Tired). AUC = 0.7985.

Biological Expert System

Knowledge-driven rules based on WHO and AAP guidelines for feeding intervals and sleep gaps.

Visual Stochastic Simulator

A 50-dimensional prototype-and-noise model to stress-test fusion before hardware deployment.

The RW-PoE Fusion Engine

The core innovation is the **Reliability-Weighted Product of Experts**:

$$\log P_{\text{fused}}(c) = \sum_{m=1}^M w_m \cdot \log P_m(c) + \text{const} \quad (1)$$



Validated Performance Metrics

Configuration	Accuracy	vs. Baseline
Audio only	56.77%	—
Audio + Biological	86.45%	+29.68 pp
Audio + Biological + Video	91.61%	+34.84 pp

Bhattacharyya Safety Barrier

Theorem 2 (Bhattacharyya Barrier)

The maximum achievable AUC for any binary classifier is bounded by:

$$\text{AUC}_{\max}(i, j) \leq \Phi \left(\sqrt{2 D_B(i, j)} \right) \quad (2)$$

Justifying the necessity of multimodal fusion for safety-critical diagnostics.

Live Demonstration

```
# Run the clinical demonstration
python benchmarks/run_demo.py
```

Citation

```
@article{benarbia2026infantcrynet,
  title = {Information-theoretic limits and universal decoding
    of human infant distress},
  author = {Safia Benarbia},
  journal= {Under Review},
  year = {2026}
}
```

Acknowledgements

Conducted under ethics protocol CHU-SBA-2025-017, with the support of CHU de Sidi Bel Abbès.